

INTRODUCING

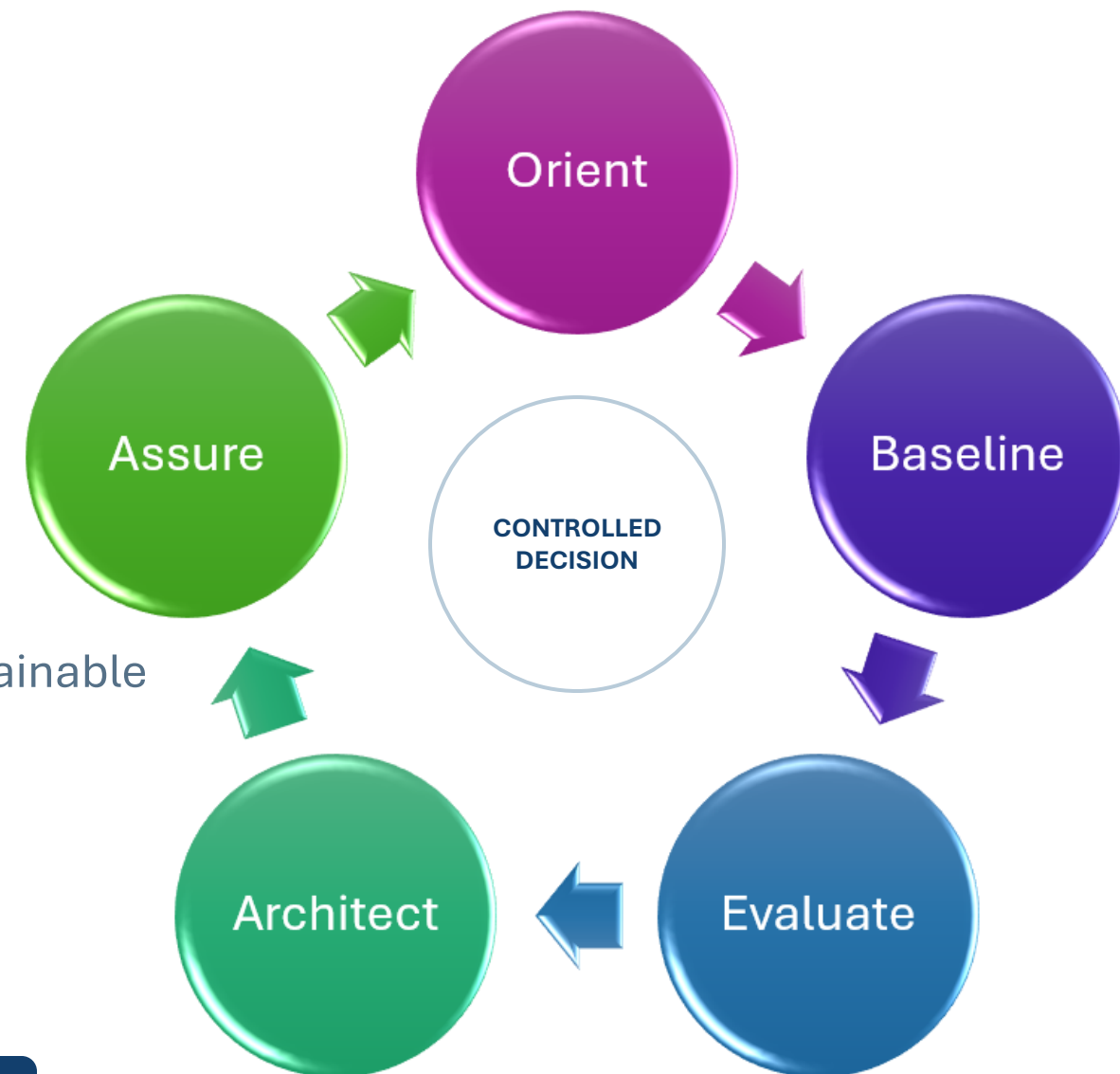
Sustainable Digital Architecture Framework

SDAF v1.0

An integrated framework for evidence-based, defensible and sustainable architectural decisions

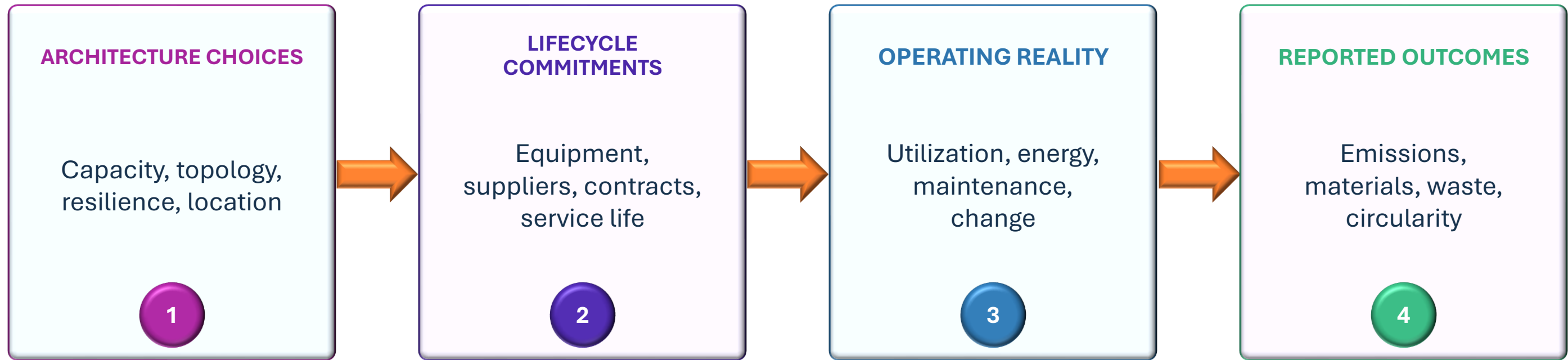
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Raymond du Toit | September 2026 | CC BY 4.0



Digital decisions create environmental consequences

Most effects are determined before they appear in sustainability reporting



Sustainability must enter while choices remain open

SDAF brings evidence into the decision process without separating sustainability from performance, resilience, security, service quality or cost.

The role of SDAF

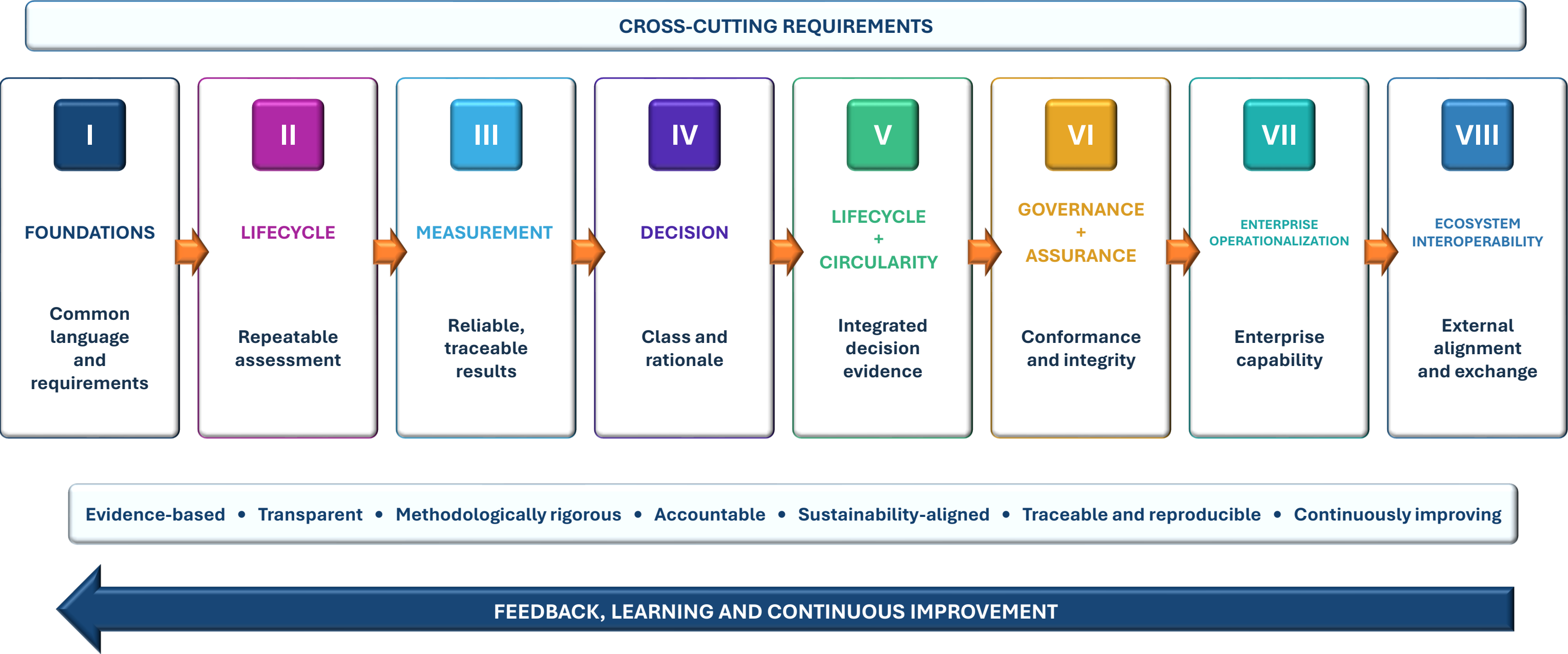
A control structure between external requirements, specialist evidence and accountable decisions



**SDAF does not replace external authorities.
It governs how their requirements and evidence inform a digital architecture decision.**

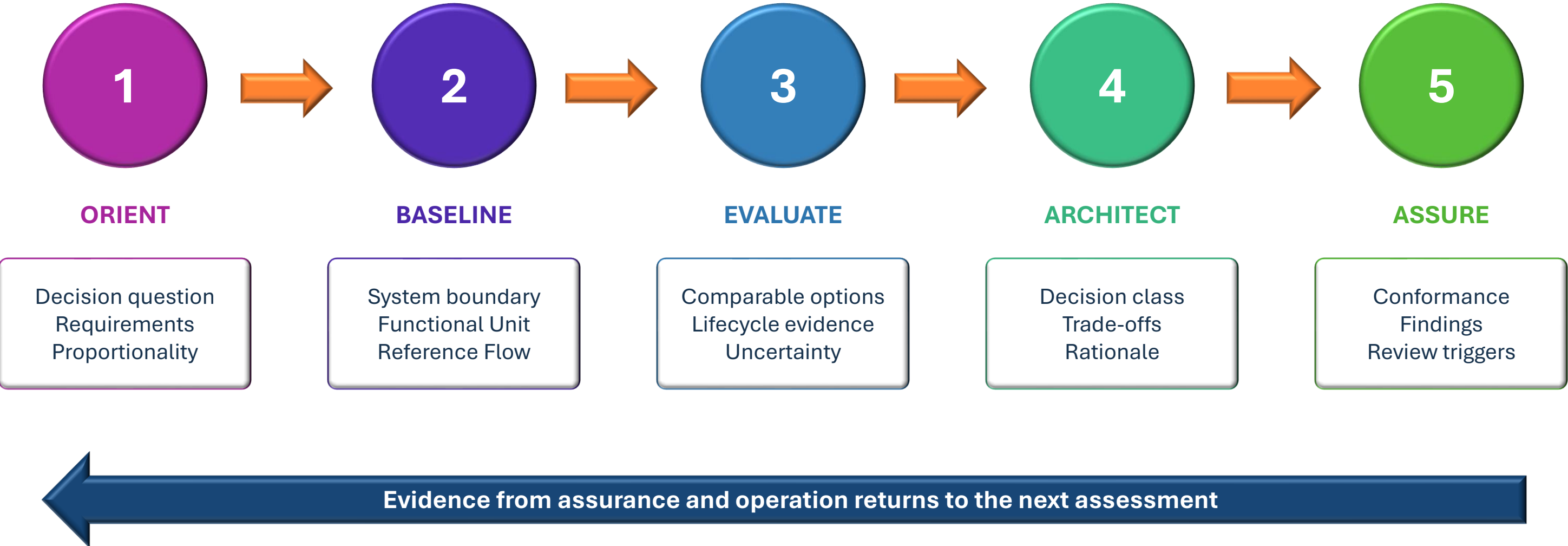
Eight Parts form one decision architecture

Parts I to V define the method and decision requirements; Parts VI to VIII enable assurance, adoption and exchange



The five-phase SDAF lifecycle

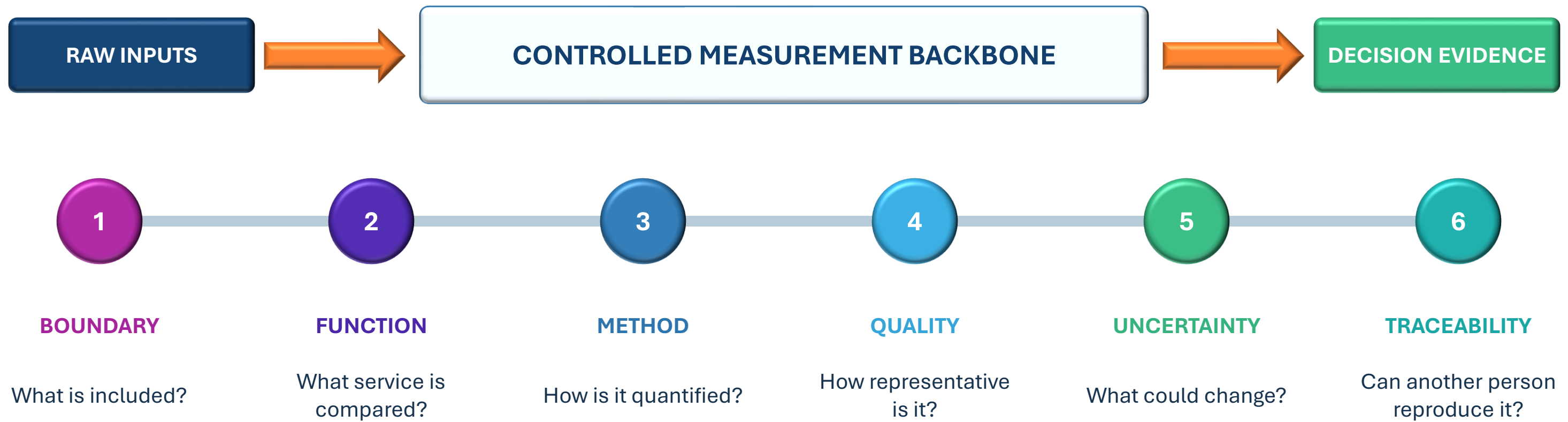
One controlled sequence carries a decision from context to verified outcome



Every phase has an entry condition, controlled activities and a required output. The sequence may iterate, but no phase disappears.

Measurement becomes decision evidence through explicit controls

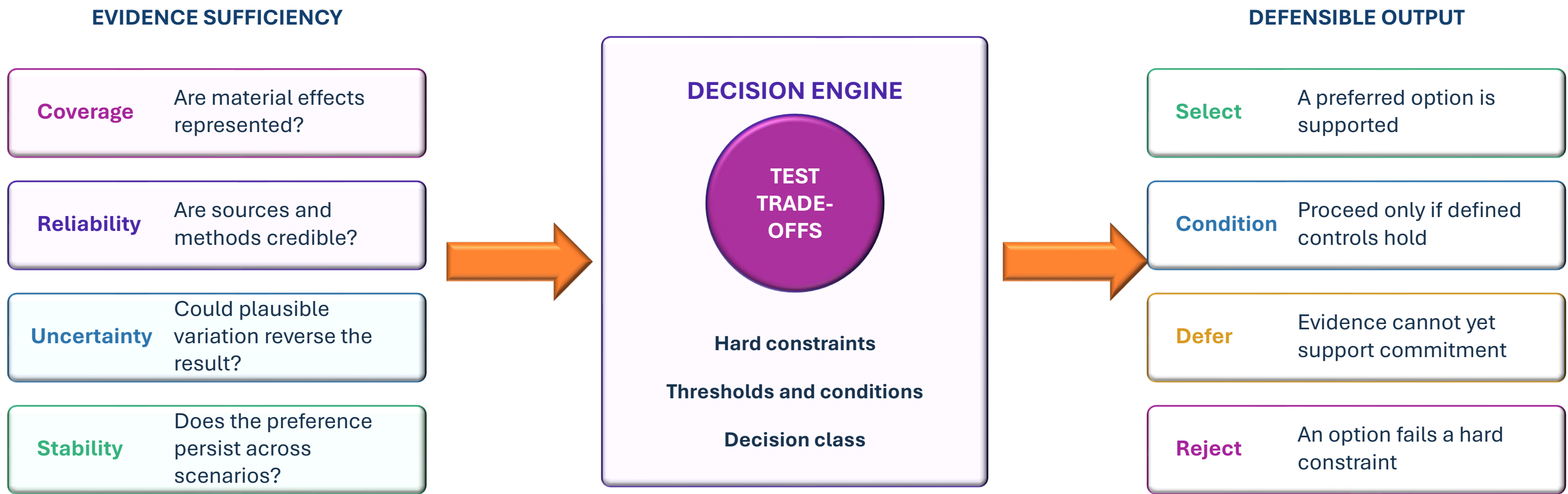
A result gains meaning only when its boundary, method, quality, uncertainty and provenance remain visible



A precise number can still be weak evidence. SDAF records confidence separately from the reported result.

Decision strength follows evidence strength and consequence

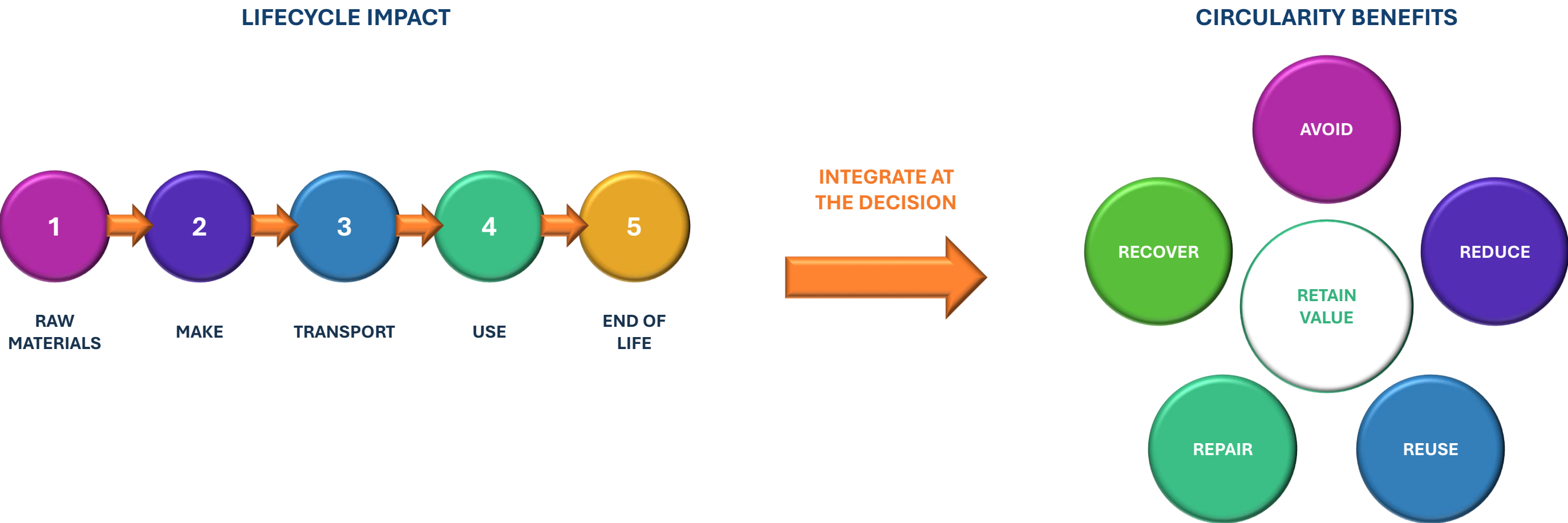
SDAF prevents a confident conclusion from resting on evidence that cannot support it



The decision record explains why the selected option remains justified, which conditions apply, and what would trigger reconsideration.

Lifecycle impact and circularity remain separate until decision integration

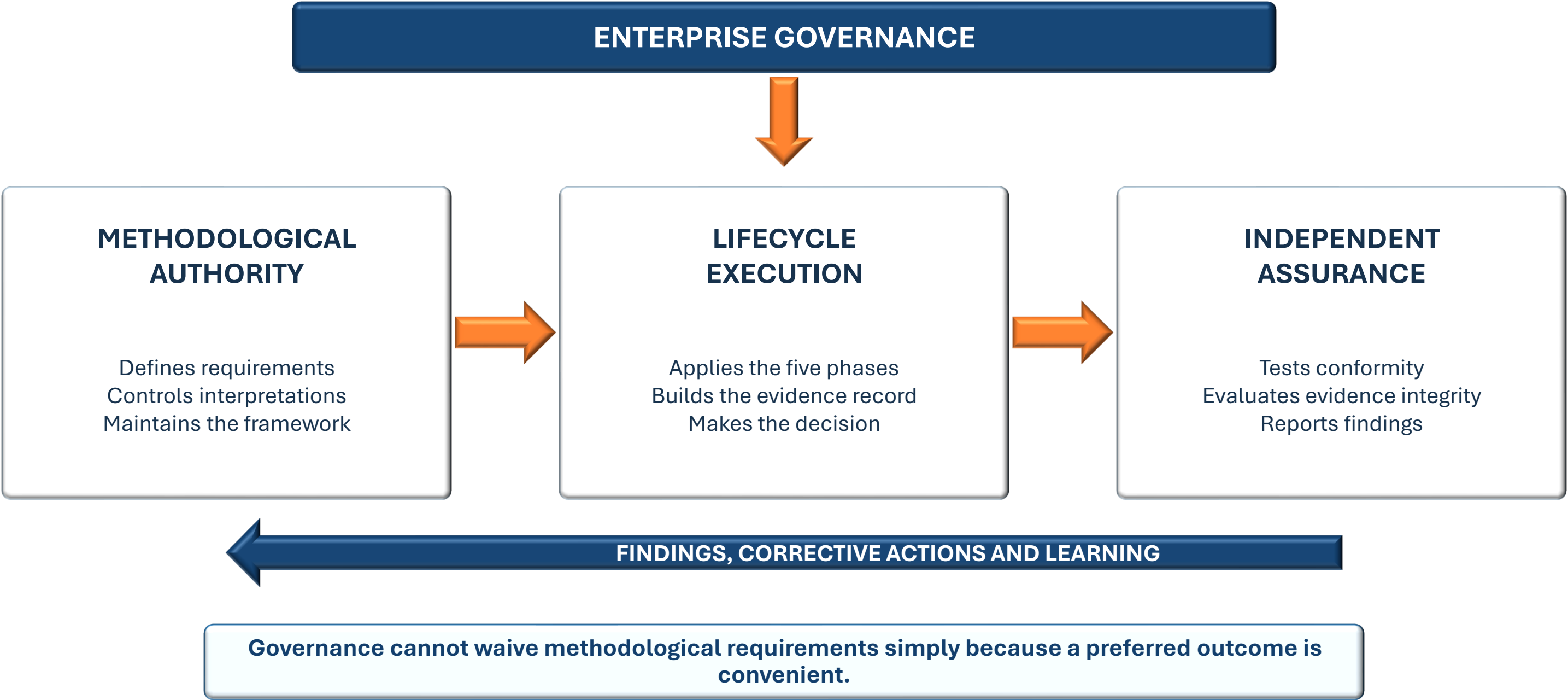
The framework preserves distinct accounting logic before comparing benefits and burdens



Keep lifecycle burdens, displaced impacts and circularity benefits traceable. State the allocation method and prevent double counting.

Governance protects the method; assurance tests its application

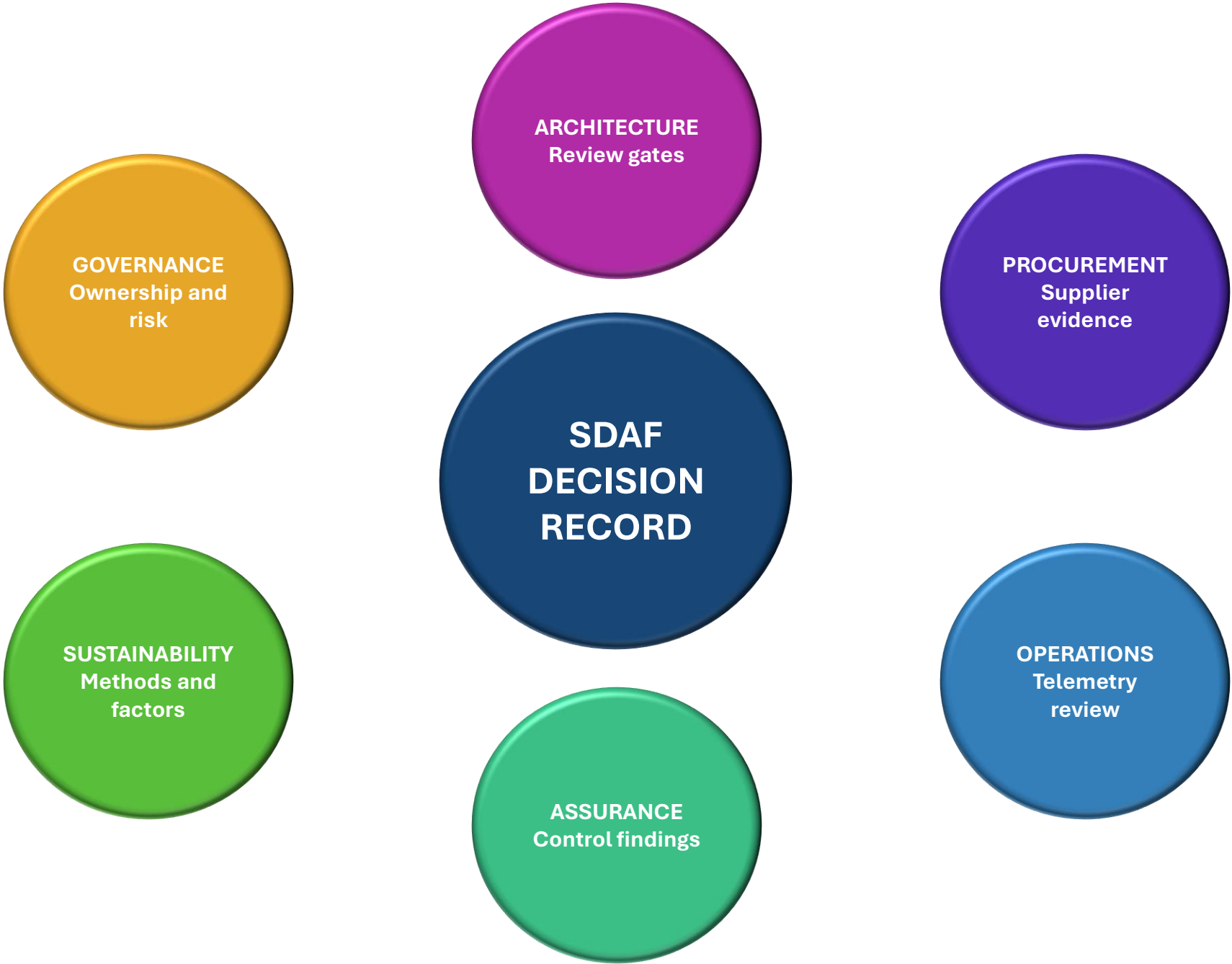
Accountability, methodological authority and independent review perform different functions



Operationalization embeds SDAF in existing enterprise decisions

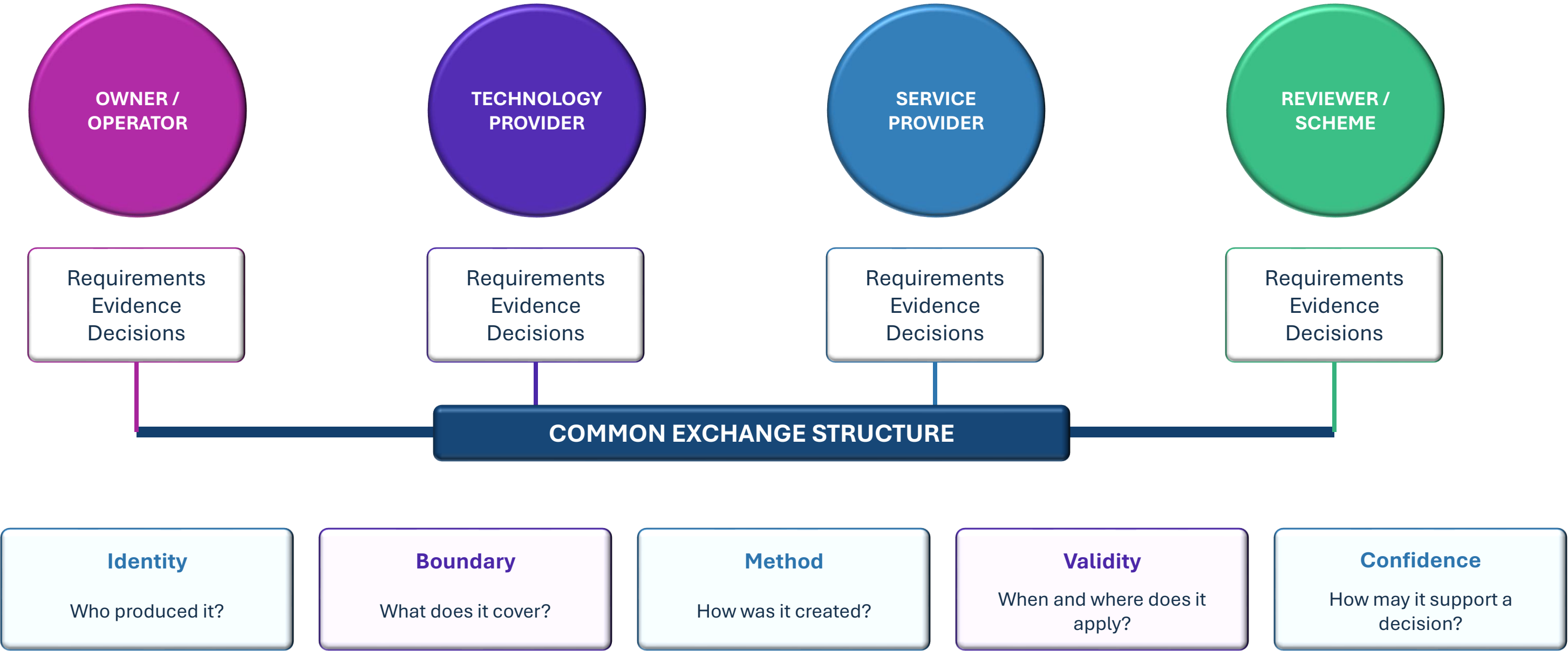
Adoption depends on roles, systems and controls rather than a parallel sustainability process

One controlled record connects existing workflows and repositories while preserving accountability and traceability.



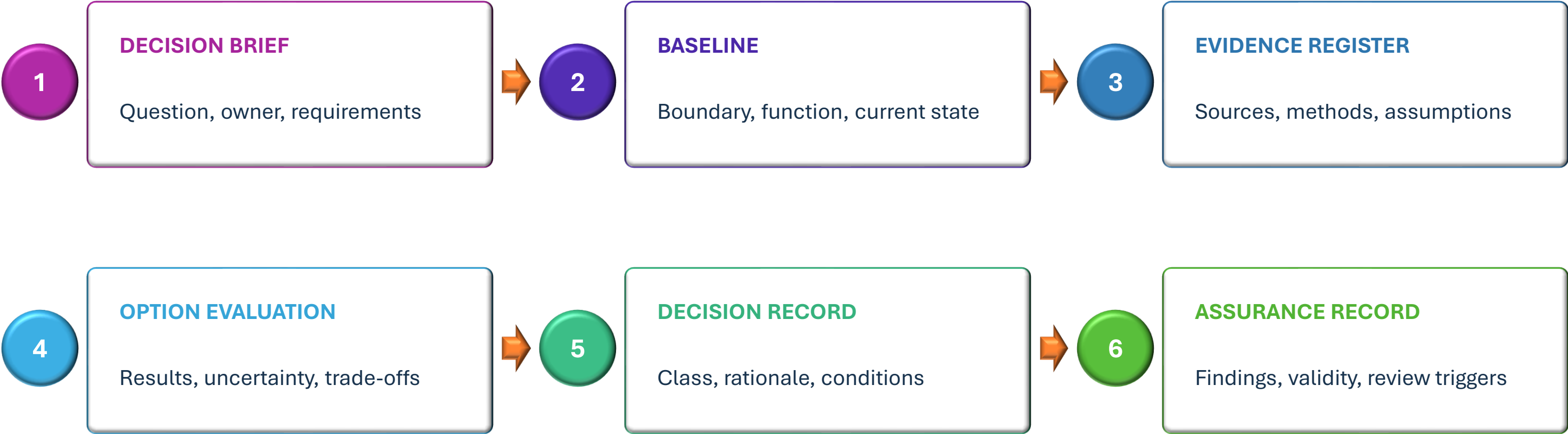
Interoperability makes evidence portable without making it equivalent

Shared structure supports exchange; provenance and authority remain attached to each record



A completed assessment leaves an auditable decision trail

The deliverable is a connected record, not a sustainability score in isolation



Together, the records show what was decided, why the evidence supported it, which limitations remain and when the conclusion must be reviewed.

Example application: retain, refurbish or replace an IT estate

SDAF changes the question from product comparison to equivalent service delivered over time

DECISION QUESTION

Which option provides the required compute service for five years while meeting resilience, security, cost and sustainability requirements?

RETAIN

Existing equipment
Maintenance and energy
Remaining useful life

REFURBISH

Selective upgrades
Extended service life
Avoided replacement

REPLACE

New equipment
Embodied impact
Operating improvement

COMMON FUNCTIONAL UNIT AND REFERENCE FLOW

Lifecycle impact
Embodied and use phase

Circularity
Life extension and recovery

Evidence confidence
Quality and uncertainty

Architecture fit
Performance and resilience

SDAF v1.0 establishes a public baseline

The framework is ready for implementation, critical review and improvement through documented experience.



USE IT

Apply SDAF to a real decision and preserve the assessment record.



TEST IT

Challenge the requirements, methods, assumptions and limits.



CONNECT IT

Map SDAF to established standards, schemes and architecture practices.



IMPROVE IT

Submit evidence from implementation and propose precise revisions.

READ AND DOWNLOAD

<https://doi.org/10.5281/zenodo.22899662>

REVIEW AND CONTRIBUTE

<https://github.com/Raymond-du-Toit/SDAF>